csv_conversion

January 24, 2026

[ ]:
```
# import os
# import pandas as pd
# import pyarrow as pa
# import pyarrow.parquet as pq
# from pathlib import Path
# import numpy as np

# base_dir = Path('E:/hneu/data')
# output_dir = Path('E:/hneu/processed_data')
# output_dir.mkdir(exist_ok=True)

# category_mapping = {
#     'fwdfinalconsumer': 'Consumer',
#     'fwdfinalenergyandultilities': 'EnergyAndUtilities',
#     'fwdfinalhealthcare': 'Healthcare',
#     'fwdfinalindustrials': 'Industrials',
#     'fwdfinalmaterials': 'Materials',
#     'fwdfinalrealestate': 'RealEstate',
#     'fwdfinaltelecompart1': 'Telecom',
#     'fwdfinaltelecompart2': 'Telecom'
# }

# def clean_numeric_value(x):
#     if pd.isna(x) or x == '' or x == 'NA' or x == 'NM':
#         return np.nan
#     if isinstance(x, str):
#         x = x.strip()
#         if x == 'NA' or x == 'NM':
#             return np.nan
#         if x.startswith('(') and x.endswith(')'):
#             x = '-' + x[1:-1]
#         x = x.replace(',', '').replace(' ', '')
#         try:
#             return float(x)
#         except:
#             return np.nan
#     return x
```

```
# for category_folder in base_dir.iterdir():
#     if not category_folder.is_dir() or 'fwdfinal' not in category_folder.name:
#         continue

#     category_name = category_mapping.get(category_folder.name, 'Unknown')
#     category_output = output_dir / category_name
#     category_output.mkdir(exist_ok=True)

#     for excel_file in category_folder.glob('*.xlsx'):
#         country_name = excel_file.stem
#         country_output = category_output / country_name
#         country_output.mkdir(exist_ok=True)

#         try:
#             df = pd.read_excel(excel_file, header=None)

#             if len(df) < 6:
#                 continue

#             col_names = df.iloc[2].values
#             field_codes = df.iloc[3].values
#             periods = df.iloc[4].values

#             columns = []
#             for i in range(len(col_names)):
#                 col_name = str(col_names[i]).strip() if not pd.
↪isna(col_names[i]) else f'col_{i}'
#                 period = str(periods[i]).strip() if not pd.isna(periods[i])␣
↪else 'Unknown'
#                 if col_name == 'nan' or col_name == '':
#                     col_name = f'col_{i}'
#                 if period == 'nan' or period == '':
#                     period = 'Unknown'
#                 columns.append(f'{col_name}_{period}')

#             data_df = df.iloc[5:].copy()
#             data_df.columns = columns

#             for col in data_df.columns:
#                 data_df[col] = data_df[col].apply(clean_numeric_value)

#             data_df = data_df[~data_df.apply(lambda row: row.astype(str).
↪isin(["NA", 'NA', "N/A", "na"]).any(), axis=1)]
#             data_df = data_df.dropna(how='all')

#             if not data_df.empty:
```

```
#                   json_path = country_output / f'{country_name}_{category_name}.
 ↪json'
#                   parquet_path = country_output /␣
 ↪f'{country_name}_{category_name}.parquet'

#                   data_df.to_json(json_path, orient='records', indent=2)

#                   table = pa.Table.from_pandas(data_df)
#                   pq.write_table(table, parquet_path)

#           except Exception as e:
#               print(f'Error processing {excel_file}: {str(e)}')
```

```
Error processing E:\hneu\data\fwdfinalconsumer\~$Argentina.xlsx: [Errno 13]
Permission denied: 'E:\\hneu\\data\\fwdfinalconsumer\\~$Argentina.xlsx'
```

[4]:
```
import duckdb

parquet_file = "E:
 ↪\hneu\processed_data\EnergyAndUtilities\Argentina\Argentina_EnergyAndUtilities.
 ↪parquet"

con = duckdb.connect()

try:
    result = con.execute(f"""
        DESCRIBE SELECT *
        FROM read_parquet('{parquet_file}')
    """).fetchdf()

    print(result.to_string())

except Exception as e:
    print(f"Error: {e}")

finally:
    con.close()
```

```
                                                 column_name column_type null
key default extra
0                                     Entity Name_Unknown      DOUBLE  YES
None    None  None
1                                       Entity ID_Unknown      DOUBLE  YES
None    None  None
2                                 Ownership Status_Unknown      DOUBLE  YES
None    None  None
3                             Industry Classification_Unknown      DOUBLE  YES
None    None  None
```

4 Gross Profit\n($000)_Latest Fiscal Year DOUBLE YES None None None
5 Interest Expense\n($000)_Latest Fiscal Year DOUBLE YES None None None
6 Net Property, Plant & Equipment\n($000)_Latest Fiscal Year DOUBLE YES None None None
7 Total Assets\n($000)_Latest Fiscal Year DOUBLE YES None None None
8 Total Debt\n($000)_Latest Fiscal Year DOUBLE YES None None None
9 Total Equity\n($000)_Latest Fiscal Year DOUBLE YES None None None
10 Capital Expenditure\n($000)_Latest Fiscal Year DOUBLE YES None None None
11 Total Revenues, 1 Year Growth\n(%)_Latest Fiscal Year DOUBLE YES None None None
12 Gross Profit Margin\n(%)_Latest Fiscal Year DOUBLE YES None None None
13 EBITDA Margin\n(%)_Latest Fiscal Year DOUBLE YES None None None
14 Net Income Margin\n(%)_Latest Fiscal Year DOUBLE YES None None None
15 Return on Assets\n(%)_Latest Fiscal Year DOUBLE YES None None None
16 Return on Capital\n(%)_Latest Fiscal Year DOUBLE YES None None None
17 Return on Equity\n(%)_Latest Fiscal Year DOUBLE YES None None None
18 Gross Profit\n($000)_FY2024 DOUBLE YES None None None
19 Gross Profit\n($000)_FY2023 DOUBLE YES None None None
20 Gross Profit\n($000)_FY2022 DOUBLE YES None None None
21 Gross Profit\n($000)_FY2021 DOUBLE YES None None None
22 Gross Profit\n($000)_FY2020 DOUBLE YES None None None
23 Gross Profit\n($000)_FY2019 DOUBLE YES None None None
24 Gross Profit\n($000)_FY2018 DOUBLE YES None None None
25 Interest Expense\n($000)_FY2024 DOUBLE YES None None None
26 Interest Expense\n($000)_FY2023 DOUBLE YES None None None
27 Interest Expense\n($000)_FY2022 DOUBLE YES None None None

28 Interest Expense\n($000)_FY2021 DOUBLE YES None None None
29 Interest Expense\n($000)_FY2020 DOUBLE YES None None None
30 Interest Expense\n($000)_FY2019 DOUBLE YES None None None
31 Interest Expense\n($000)_FY2018 DOUBLE YES None None None
32 Net Property, Plant & Equipment\n($000)_FY2024 DOUBLE YES None None None
33 Net Property, Plant & Equipment\n($000)_FY2023 DOUBLE YES None None None
34 Net Property, Plant & Equipment\n($000)_FY2022 DOUBLE YES None None None
35 Net Property, Plant & Equipment\n($000)_FY2021 DOUBLE YES None None None
36 Net Property, Plant & Equipment\n($000)_FY2020 DOUBLE YES None None None
37 Net Property, Plant & Equipment\n($000)_FY2019 DOUBLE YES None None None
38 Net Property, Plant & Equipment\n($000)_FY2018 DOUBLE YES None None None
39 Total Assets\n($000)_FY2024 DOUBLE YES None None None
40 Total Assets\n($000)_FY2023 DOUBLE YES None None None
41 Total Assets\n($000)_FY2022 DOUBLE YES None None None
42 Total Assets\n($000)_FY2021 DOUBLE YES None None None
43 Total Assets\n($000)_FY2020 DOUBLE YES None None None
44 Total Assets\n($000)_FY2019 DOUBLE YES None None None
45 Total Assets\n($000)_FY2018 DOUBLE YES None None None
46 Total Debt\n($000)_FY2024 DOUBLE YES None None None
47 Total Debt\n($000)_FY2023 DOUBLE YES None None None
48 Total Debt\n($000)_FY2022 DOUBLE YES None None None
49 Total Debt\n($000)_FY2021 DOUBLE YES None None None
50 Total Debt\n($000)_FY2020 DOUBLE YES None None None
51 Total Debt\n($000)_FY2019 DOUBLE YES None None None

```
52                                  Total Debt\n($000)_FY2018      DOUBLE  YES
None    None  None
53                                Total Equity\n($000)_FY2024      DOUBLE  YES
None    None  None
54                                Total Equity\n($000)_FY2023      DOUBLE  YES
None    None  None
55                                Total Equity\n($000)_FY2022      DOUBLE  YES
None    None  None
56                                Total Equity\n($000)_FY2021      DOUBLE  YES
None    None  None
57                                Total Equity\n($000)_FY2020      DOUBLE  YES
None    None  None
58                                Total Equity\n($000)_FY2019      DOUBLE  YES
None    None  None
59                                Total Equity\n($000)_FY2018      DOUBLE  YES
None    None  None
60                          Capital Expenditure\n($000)_FY2024      DOUBLE  YES
None    None  None
61                          Capital Expenditure\n($000)_FY2023      DOUBLE  YES
None    None  None
62                          Capital Expenditure\n($000)_FY2022      DOUBLE  YES
None    None  None
63                          Capital Expenditure\n($000)_FY2021      DOUBLE  YES
None    None  None
64                          Capital Expenditure\n($000)_FY2020      DOUBLE  YES
None    None  None
65                          Capital Expenditure\n($000)_FY2019      DOUBLE  YES
None    None  None
66                          Capital Expenditure\n($000)_FY2018      DOUBLE  YES
None    None  None
67                    Total Revenues, 1 Year Growth\n(%)_FY2024      DOUBLE  YES
None    None  None
68                    Total Revenues, 1 Year Growth\n(%)_FY2023      DOUBLE  YES
None    None  None
69                    Total Revenues, 1 Year Growth\n(%)_FY2022      DOUBLE  YES
None    None  None
70                    Total Revenues, 1 Year Growth\n(%)_FY2021      DOUBLE  YES
None    None  None
71                    Total Revenues, 1 Year Growth\n(%)_FY2020      DOUBLE  YES
None    None  None
72                    Total Revenues, 1 Year Growth\n(%)_FY2019      DOUBLE  YES
None    None  None
73                    Total Revenues, 1 Year Growth\n(%)_FY2018      DOUBLE  YES
None    None  None
74                              Gross Profit Margin\n(%)_FY2024      DOUBLE  YES
None    None  None
75                              Gross Profit Margin\n(%)_FY2023      DOUBLE  YES
None    None  None
```

76 Gross Profit Margin\n(%)_FY2022 DOUBLE YES
None None None
77 Gross Profit Margin\n(%)_FY2021 DOUBLE YES
None None None
78 Gross Profit Margin\n(%)_FY2020 DOUBLE YES
None None None
79 Gross Profit Margin\n(%)_FY2019 DOUBLE YES
None None None
80 Gross Profit Margin\n(%)_FY2018 DOUBLE YES
None None None
81 EBITDA Margin\n(%)_FY2024 DOUBLE YES
None None None
82 EBITDA Margin\n(%)_FY2023 DOUBLE YES
None None None
83 EBITDA Margin\n(%)_FY2022 DOUBLE YES
None None None
84 EBITDA Margin\n(%)_FY2021 DOUBLE YES
None None None
85 EBITDA Margin\n(%)_FY2020 DOUBLE YES
None None None
86 EBITDA Margin\n(%)_FY2019 DOUBLE YES
None None None
87 EBITDA Margin\n(%)_FY2018 DOUBLE YES
None None None
88 Net Income Margin\n(%)_FY2024 DOUBLE YES
None None None
89 Net Income Margin\n(%)_FY2023 DOUBLE YES
None None None
90 Net Income Margin\n(%)_FY2022 DOUBLE YES
None None None
91 Net Income Margin\n(%)_FY2021 DOUBLE YES
None None None
92 Net Income Margin\n(%)_FY2020 DOUBLE YES
None None None
93 Net Income Margin\n(%)_FY2019 DOUBLE YES
None None None
94 Net Income Margin\n(%)_FY2018 DOUBLE YES
None None None
95 Return on Assets\n(%)_FY2024 DOUBLE YES
None None None
96 Return on Assets\n(%)_FY2023 DOUBLE YES
None None None
97 Return on Assets\n(%)_FY2022 DOUBLE YES
None None None
98 Return on Assets\n(%)_FY2021 DOUBLE YES
None None None
99 Return on Assets\n(%)_FY2020 DOUBLE YES
None None None

```
100                                Return on Assets\n(%)_FY2019      DOUBLE  YES
None    None  None
101                                Return on Assets\n(%)_FY2018      DOUBLE  YES
None    None  None
102                               Return on Capital\n(%)_FY2024      DOUBLE  YES
None    None  None
103                               Return on Capital\n(%)_FY2023      DOUBLE  YES
None    None  None
104                               Return on Capital\n(%)_FY2022      DOUBLE  YES
None    None  None
105                               Return on Capital\n(%)_FY2021      DOUBLE  YES
None    None  None
106                               Return on Capital\n(%)_FY2020      DOUBLE  YES
None    None  None
107                               Return on Capital\n(%)_FY2019      DOUBLE  YES
None    None  None
108                               Return on Capital\n(%)_FY2018      DOUBLE  YES
None    None  None
109                                Return on Equity\n(%)_FY2024      DOUBLE  YES
None    None  None
110                                Return on Equity\n(%)_FY2023      DOUBLE  YES
None    None  None
111                                Return on Equity\n(%)_FY2022      DOUBLE  YES
None    None  None
112                                Return on Equity\n(%)_FY2021      DOUBLE  YES
None    None  None
113                                Return on Equity\n(%)_FY2020      DOUBLE  YES
None    None  None
114                                Return on Equity\n(%)_FY2019      DOUBLE  YES
None    None  None
115                                Return on Equity\n(%)_FY2018      DOUBLE  YES
None    None  None
116                                            Geography_Unknown      DOUBLE  YES
None    None  None
```

[ ]:

```python

```

[1]:

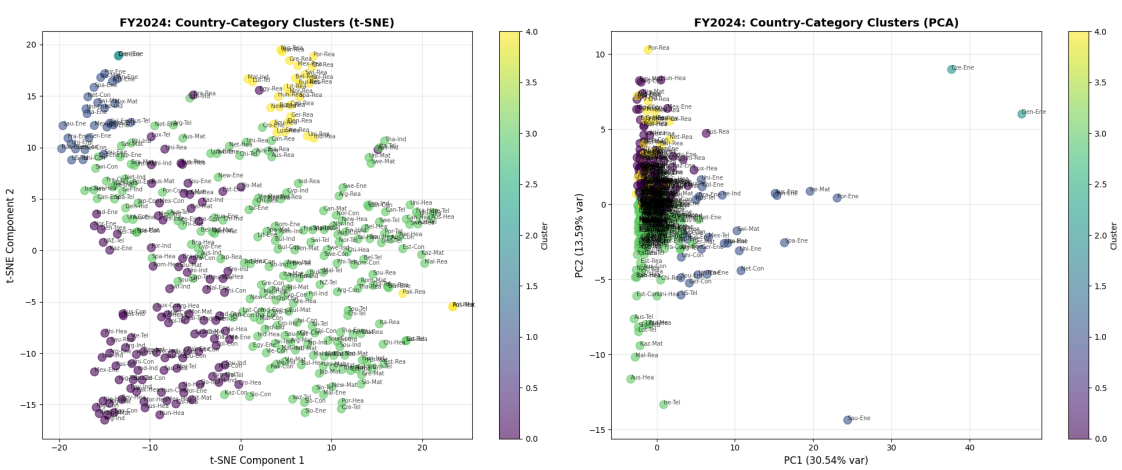
```python
import pandas as pd
import numpy as np
from pathlib import Path
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans
from sklearn.manifold import TSNE
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
```

```
base_dir = Path('E:/hneu/processed_data')

fiscal_years = ['FY2024', 'FY2023', 'FY2022', 'FY2021', 'FY2020', 'FY2019',␣
↪'FY2018', 'Latest Fiscal Year']

indicator_base_names = [
    'Gross Profit Margin\n(%)',
    'EBITDA Margin\n(%)',
    'Net Income Margin\n(%)',
    'Return on Assets\n(%)',
    'Return on Capital\n(%)',
    'Return on Equity\n(%)',
    'Total Revenues, 1 Year Growth\n(%)',
    'Gross Profit\n($000)',
    'Interest Expense\n($000)',
    'Net Property, Plant & Equipment\n($000)',
    'Total Assets\n($000)',
    'Total Debt\n($000)',
    'Total Equity\n($000)',
    'Capital Expenditure\n($000)'
]

all_data = []

for category_dir in base_dir.iterdir():
    if not category_dir.is_dir():
        continue

    category_name = category_dir.name

    for country_dir in category_dir.iterdir():
        if not country_dir.is_dir():
            continue

        country_name = country_dir.name

        for parquet_file in country_dir.glob('*.parquet'):
            df = pd.read_parquet(parquet_file)

            for fy in fiscal_years:
                fy_label = 'FY2025' if fy == 'Latest Fiscal Year' else fy

                indicator_cols = [f'{ind}_{fy}' for ind in indicator_base_names]
                available_cols = [col for col in indicator_cols if col in df.
↪columns]
```

```
                if len(available_cols) < len(indicator_base_names):
                    continue

                df_fy = df[available_cols].copy()
                df_fy = df_fy.dropna()

                if len(df_fy) == 0:
                    continue

                aggregated_values = []

                for col in available_cols:
                    values = df_fy[col].values

                    if len(values) > 0:
                        aggregated_values.extend([
                            np.mean(values),
                            np.std(values),
                            np.median(values),
                            np.percentile(values, 25),
                            np.percentile(values, 75)
                        ])
                    else:
                        aggregated_values.extend([np.nan] * 5)

                all_data.append({
                    'Category': category_name,
                    'Country': country_name,
                    'FiscalYear': fy_label,
                    'Vector': aggregated_values,
                    'NumCompanies': len(df_fy)
                })

df_aggregated = pd.DataFrame(all_data)

vector_df = pd.DataFrame(df_aggregated['Vector'].tolist())
df_final = pd.concat([df_aggregated[['Category', 'Country', 'FiscalYear',␣
 ↪'NumCompanies']], vector_df], axis=1)

df_final = df_final.dropna()

df_fy2024 = df_final[df_final['FiscalYear'] == 'FY2024'].copy()

if len(df_fy2024) > 0:
    X = df_fy2024.iloc[:, 4:].values

    scaler = StandardScaler()
```

```
X_scaled = scaler.fit_transform(X)

n_components = min(10, X_scaled.shape[1], X_scaled.shape[0])
pca = PCA(n_components=n_components)
X_pca = pca.fit_transform(X_scaled)

optimal_k = min(5, len(df_fy2024) - 1)
kmeans = KMeans(n_clusters=optimal_k, random_state=42, n_init=10)
df_fy2024['Cluster'] = kmeans.fit_predict(X_pca)

tsne = TSNE(n_components=2, random_state=42, perplexity=min(30,␣
↪len(df_fy2024) - 1))
X_tsne = tsne.fit_transform(X_pca)

fig, axes = plt.subplots(1, 2, figsize=(20, 8))

scatter = axes[0].scatter(X_tsne[:, 0], X_tsne[:, 1],
                          c=df_fy2024['Cluster'],
                          cmap='viridis',
                          s=100,
                          alpha=0.6)

for idx, row in df_fy2024.iterrows():
    axes[0].annotate(f"{row['Country'][:3]}-{row['Category'][:3]}",
                     (X_tsne[df_fy2024.index.get_loc(idx), 0],
                      X_tsne[df_fy2024.index.get_loc(idx), 1]),
                     fontsize=7, alpha=0.7)

axes[0].set_xlabel('t-SNE Component 1', fontsize=12)
axes[0].set_ylabel('t-SNE Component 2', fontsize=12)
axes[0].set_title('FY2024: Country-Category Clusters (t-SNE)', fontsize=14,␣
↪fontweight='bold')
axes[0].grid(True, alpha=0.3)
plt.colorbar(scatter, ax=axes[0], label='Cluster')

pca_2d = PCA(n_components=2)
X_pca_2d = pca_2d.fit_transform(X_scaled)

scatter2 = axes[1].scatter(X_pca_2d[:, 0], X_pca_2d[:, 1],
                           c=df_fy2024['Cluster'],
                           cmap='viridis',
                           s=100,
                           alpha=0.6)

for idx, row in df_fy2024.iterrows():
    axes[1].annotate(f"{row['Country'][:3]}-{row['Category'][:3]}",
                     (X_pca_2d[df_fy2024.index.get_loc(idx), 0],
```

```
                                X_pca_2d[df_fy2024.index.get_loc(idx), 1]),
                               fontsize=7, alpha=0.7)

    axes[1].set_xlabel(f'PC1 ({pca_2d.explained_variance_ratio_[0]:.2%} var)',␣
↪fontsize=12)
    axes[1].set_ylabel(f'PC2 ({pca_2d.explained_variance_ratio_[1]:.2%} var)',␣
↪fontsize=12)
    axes[1].set_title('FY2024: Country-Category Clusters (PCA)', fontsize=14,␣
↪fontweight='bold')
    axes[1].grid(True, alpha=0.3)
    plt.colorbar(scatter2, ax=axes[1], label='Cluster')

    plt.tight_layout()
    plt.savefig('fy2024_clustering.png', dpi=300, bbox_inches='tight')
    plt.show()

    print(f"Total country-category pairs in FY2024: {len(df_fy2024)}")
    print(f"Number of clusters: {optimal_k}")
    print(f"\nPCA explained variance ratio (top {n_components} components):")
    print(pca.explained_variance_ratio_)
    print(f"\nCumulative variance explained: {pca.explained_variance_ratio_.
↪sum():.2%}")

    print("\nCluster distribution:")
    print(df_fy2024.groupby('Cluster')[['Category', 'Country']].apply(lambda x:␣
↪list(zip(x['Country'], x['Category']))))

df_final.to_csv('country_category_vectors.csv', index=False)
print("\nData saved to country_category_vectors.csv")
```




```
Total country-category pairs in FY2024: 367
```

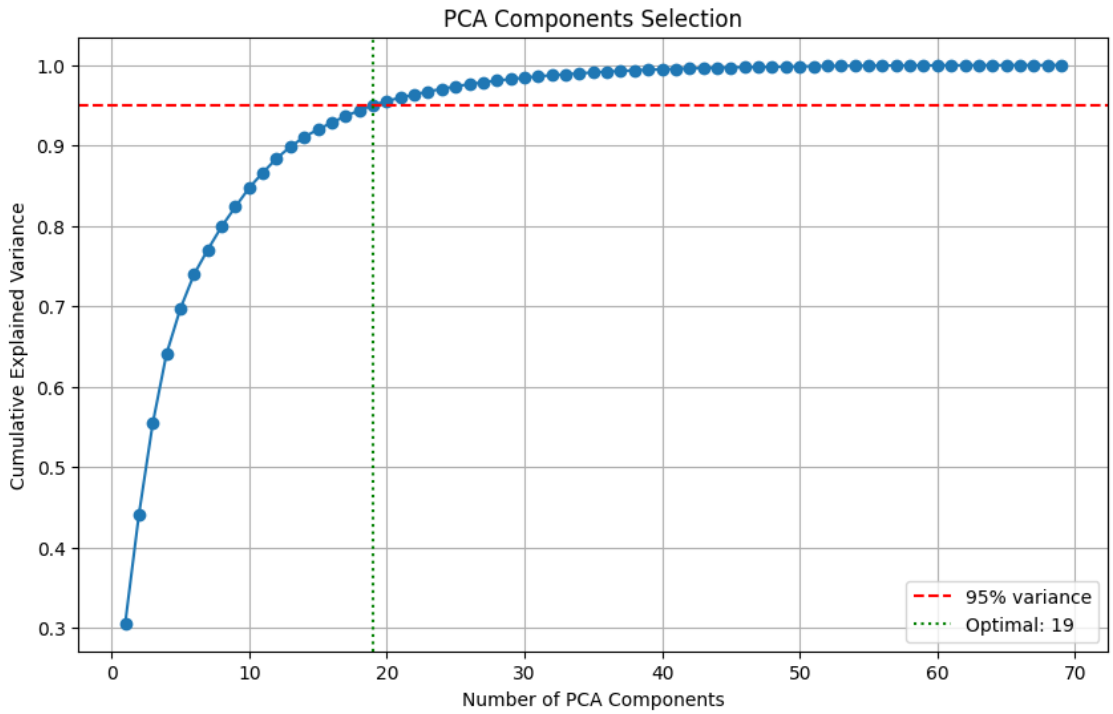
```
Number of clusters: 5

PCA explained variance ratio (top 10 components):
[0.30539386 0.13589204 0.11285311 0.08647035 0.05591195 0.04306895
 0.03077676 0.02857541 0.02453275 0.02366315]

Cumulative variance explained: 84.71%

Cluster distribution:
Cluster
0    [(Brazil, Consumer), (Colombia, Consumer), (Cz…
1    [(Belgium, Consumer), (Germany, Consumer), (Ne…
2    [(Czechia, EnergyAndUtilities), (Denmark, Ener…
3    [(Argentina, Consumer), (Australia, Consumer),…
4    [(Poland, Healthcare), (Malta, Industrials), (…
dtype: object

Data saved to country_category_vectors.csv
```

[1]:
```
import numpy as np
import pandas as pd
from sklearn.metrics import silhouette_score, calinski_harabasz_score,␣
 ↪davies_bouldin_score
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
import matplotlib.pyplot as plt

df_fy2024 = pd.read_csv('country_category_vectors.csv')
df_fy2024 = df_fy2024[df_fy2024['FiscalYear'] == 'FY2024']
df_fy2024 = df_fy2024.dropna()

X_original = df_fy2024.iloc[:, 4:].values
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_original)

n_samples = X_scaled.shape[0]
n_features = X_scaled.shape[1]

explained_variances = []
for n in range(1, min(n_features, n_samples)):
    pca = PCA(n_components=n)
    X_pca = pca.fit_transform(X_scaled)
    explained_variances.append(np.sum(pca.explained_variance_ratio_))

optimal_pca_idx = np.argmax(np.array(explained_variances) >= 0.95)
```

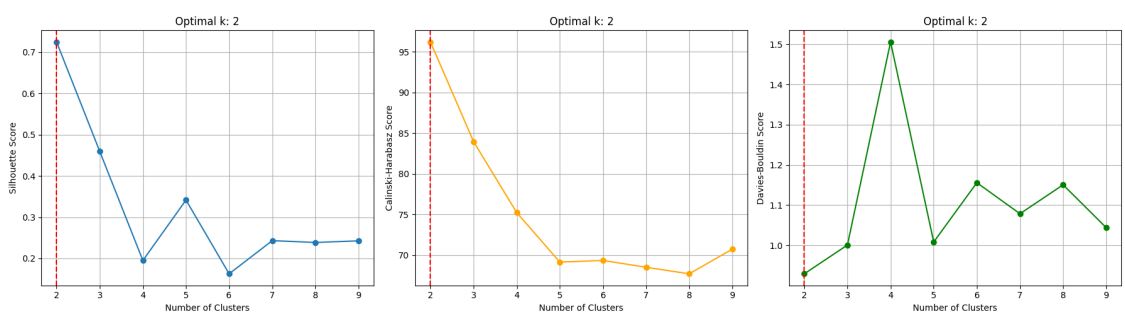
```
optimal_pca_components = optimal_pca_idx + 1 if optimal_pca_idx <␣
↪len(explained_variances) else len(explained_variances)

plt.figure(figsize=(10, 6))
plt.plot(range(1, len(explained_variances) + 1), explained_variances,␣
↪marker='o')
plt.axhline(y=0.95, color='r', linestyle='--', label='95% variance')
plt.axvline(x=optimal_pca_components, color='g', linestyle=':', label=f'Optimal:
↪ {optimal_pca_components}')
plt.xlabel('Number of PCA Components')
plt.ylabel('Cumulative Explained Variance')
plt.title('PCA Components Selection')
plt.legend()
plt.grid(True)
plt.savefig('pca_variance.png')
plt.show()

print(f"Optimal PCA components: {optimal_pca_components}")

pca_optimal = PCA(n_components=optimal_pca_components)
X_pca_optimal = pca_optimal.fit_transform(X_scaled)

silhouette_scores = []
ch_scores = []
db_scores = []
k_range = range(2, min(10, n_samples))

for k in k_range:
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    labels = kmeans.fit_predict(X_pca_optimal)

    if len(np.unique(labels)) > 1:
        silhouette_scores.append(silhouette_score(X_pca_optimal, labels))
        ch_scores.append(calinski_harabasz_score(X_pca_optimal, labels))
        db_scores.append(davies_bouldin_score(X_pca_optimal, labels))
    else:
        silhouette_scores.append(0)
        ch_scores.append(0)
        db_scores.append(np.inf)

optimal_k_silhouette = k_range[np.argmax(silhouette_scores)] if␣
↪silhouette_scores else 2
optimal_k_ch = k_range[np.argmax(ch_scores)] if ch_scores else 2
optimal_k_db = k_range[np.argmin(db_scores)] if db_scores else 2

fig, axes = plt.subplots(1, 3, figsize=(18, 5))
```

```
axes[0].plot(k_range, silhouette_scores, marker='o')
axes[0].axvline(x=optimal_k_silhouette, color='r', linestyle='--')
axes[0].set_xlabel('Number of Clusters')
axes[0].set_ylabel('Silhouette Score')
axes[0].set_title(f'Optimal k: {optimal_k_silhouette}')
axes[0].grid(True)

axes[1].plot(k_range, ch_scores, marker='o', color='orange')
axes[1].axvline(x=optimal_k_ch, color='r', linestyle='--')
axes[1].set_xlabel('Number of Clusters')
axes[1].set_ylabel('Calinski-Harabasz Score')
axes[1].set_title(f'Optimal k: {optimal_k_ch}')
axes[1].grid(True)

axes[2].plot(k_range, db_scores, marker='o', color='green')
axes[2].axvline(x=optimal_k_db, color='r', linestyle='--')
axes[2].set_xlabel('Number of Clusters')
axes[2].set_ylabel('Davies-Bouldin Score')
axes[2].set_title(f'Optimal k: {optimal_k_db}')
axes[2].grid(True)

plt.tight_layout()
plt.savefig('cluster_metrics.png')
plt.show()

kmeans_final = KMeans(n_clusters=optimal_k_silhouette, random_state=42,␣
 ↪n_init=10)
final_labels = kmeans_final.fit_predict(X_pca_optimal)

silhouette_final = silhouette_score(X_pca_optimal, final_labels)
ch_final = calinski_harabasz_score(X_pca_optimal, final_labels)
db_final = davies_bouldin_score(X_pca_optimal, final_labels)

print(f"Final Clustering Results:")
print(f"Number of clusters: {optimal_k_silhouette}")
print(f"PCA components: {optimal_pca_components}")
print(f"Silhouette Score: {silhouette_final:.4f}")
print(f"Calinski-Harabasz Score: {ch_final:.4f}")
print(f"Davies-Bouldin Score: {db_final:.4f}")
print(f"\nCluster sizes:")
unique, counts = np.unique(final_labels, return_counts=True)
for cluster, count in zip(unique, counts):
    print(f"Cluster {cluster}: {count} samples")

cluster_stats = pd.DataFrame({
    'Cluster': final_labels,
    'Country': df_fy2024['Country'].values,
```

```
    'Category': df_fy2024['Category'].values
})
cluster_stats.to_csv('cluster_assignments.csv', index=False)
```




```
Optimal PCA components: 19
```




```
Final Clustering Results:
Number of clusters: 2
PCA components: 19
Silhouette Score: 0.7247
Calinski-Harabasz Score: 96.1957
Davies-Bouldin Score: 0.9291
```

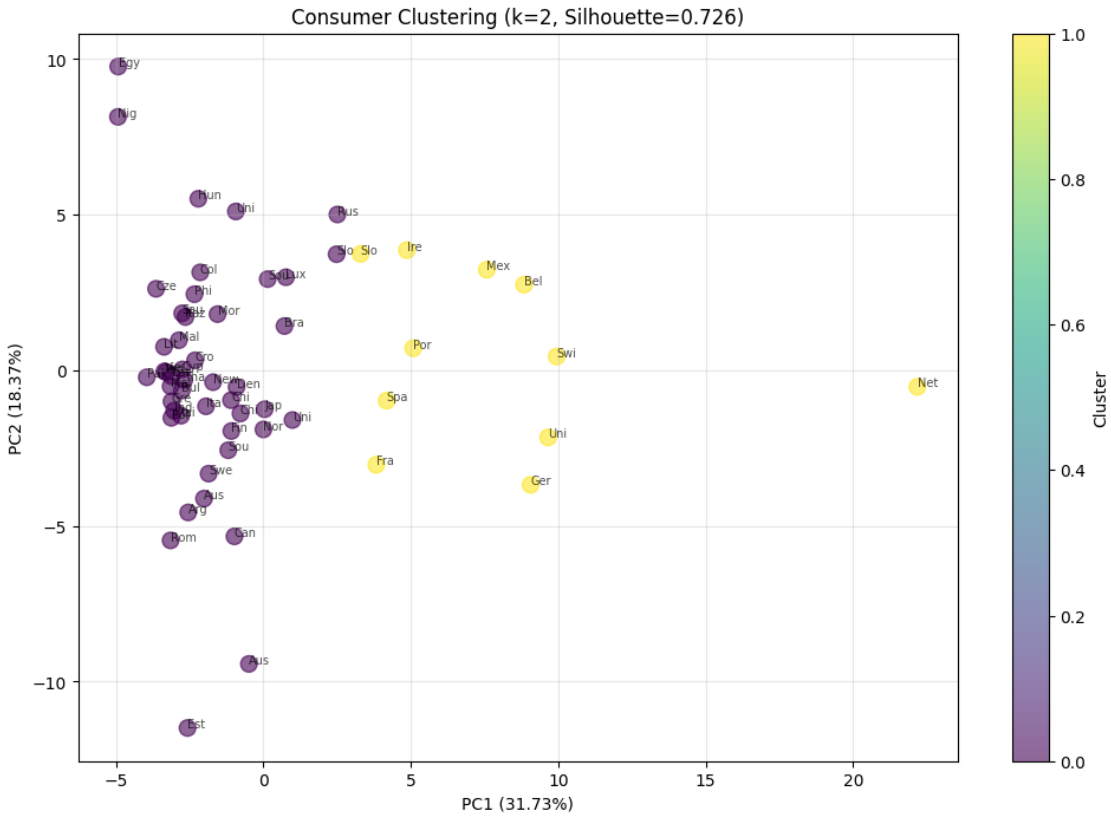
```
Cluster sizes:
Cluster 0: 359 samples
Cluster 1: 8 samples
```

[2]:
```python
import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score, calinski_harabasz_score,␣
↪davies_bouldin_score
import matplotlib.pyplot as plt

df = pd.read_csv('country_category_vectors.csv')
df_fy2024 = df[df['FiscalYear'] == 'FY2024']
df_fy2024 = df_fy2024.dropna()

categories = df_fy2024['Category'].unique()
all_metrics = []

for category in categories:
    df_category = df_fy2024[df_fy2024['Category'] == category]

    if len(df_category) < 2:
        continue

    X = df_category.iloc[:, 4:].values
    scaler = StandardScaler()
    X_scaled = scaler.fit_transform(X)

    n_components_range = range(1, min(10, X_scaled.shape[0], X_scaled.shape[1]))
    explained_variances = []

    for n in n_components_range:
        pca = PCA(n_components=n)
        X_pca = pca.fit_transform(X_scaled)
        explained_variances.append(np.sum(pca.explained_variance_ratio_))

    optimal_pca_idx = np.argmax(np.array(explained_variances) >= 0.95)
    optimal_pca_components = optimal_pca_idx + 1 if optimal_pca_idx <␣
↪len(explained_variances) else len(explained_variances)

    pca_optimal = PCA(n_components=optimal_pca_components)
    X_pca_optimal = pca_optimal.fit_transform(X_scaled)

    silhouette_scores = []
    ch_scores = []
```

```
db_scores = []
k_range = range(2, min(10, len(df_category)))

for k in k_range:
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    labels = kmeans.fit_predict(X_pca_optimal)

    if len(np.unique(labels)) > 1:
        silhouette_scores.append(silhouette_score(X_pca_optimal, labels))
        ch_scores.append(calinski_harabasz_score(X_pca_optimal, labels))
        db_scores.append(davies_bouldin_score(X_pca_optimal, labels))
    else:
        silhouette_scores.append(0)
        ch_scores.append(0)
        db_scores.append(np.inf)

optimal_k = k_range[np.argmax(silhouette_scores)] if silhouette_scores else␣
↪2

kmeans_final = KMeans(n_clusters=optimal_k, random_state=42, n_init=10)
final_labels = kmeans_final.fit_predict(X_pca_optimal)

silhouette_final = silhouette_score(X_pca_optimal, final_labels) if len(np.
↪unique(final_labels)) > 1 else 0
ch_final = calinski_harabasz_score(X_pca_optimal, final_labels) if len(np.
↪unique(final_labels)) > 1 else 0
db_final = davies_bouldin_score(X_pca_optimal, final_labels) if len(np.
↪unique(final_labels)) > 1 else np.inf

pca_2d = PCA(n_components=2)
X_pca_2d = pca_2d.fit_transform(X_scaled)

plt.figure(figsize=(12, 8))
scatter = plt.scatter(X_pca_2d[:, 0], X_pca_2d[:, 1], c=final_labels,␣
↪cmap='viridis', s=100, alpha=0.6)

for idx, row in df_category.iterrows():
    plt.annotate(row['Country'][:3], (X_pca_2d[df_category.index.
↪get_loc(idx), 0],
                                      X_pca_2d[df_category.index.
↪get_loc(idx), 1]),
                 fontsize=7, alpha=0.7)

plt.xlabel(f'PC1 ({pca_2d.explained_variance_ratio_[0]:.2%})')
plt.ylabel(f'PC2 ({pca_2d.explained_variance_ratio_[1]:.2%})')
```

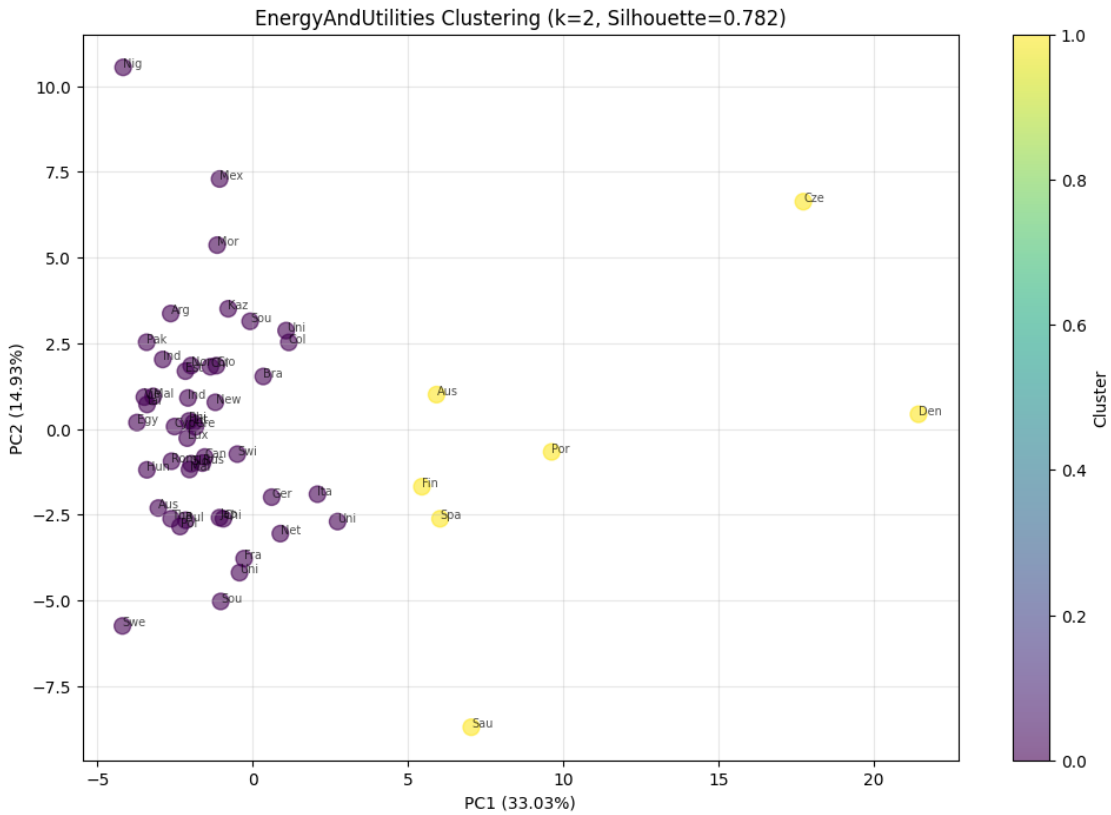
```
    plt.title(f'{category} Clustering (k={optimal_k},␣
↪Silhouette={silhouette_final:.3f})')
    plt.colorbar(scatter, label='Cluster')
    plt.grid(True, alpha=0.3)
    plt.savefig(f'clustering_{category.replace(" ", "_")}.png', dpi=300,␣
↪bbox_inches='tight')
    plt.show()

    unique, counts = np.unique(final_labels, return_counts=True)
    cluster_distribution = dict(zip(unique, counts))

    all_metrics.append({
        'Category': category,
        'Samples': len(df_category),
        'Optimal_PCA': optimal_pca_components,
        'Optimal_Clusters': optimal_k,
        'Silhouette': silhouette_final,
        'Calinski_Harabasz': ch_final,
        'Davies_Bouldin': db_final,
        'Cluster_Distribution': cluster_distribution
    })

    print(f"\n=== {category} ===")
    print(f"Samples: {len(df_category)}")
    print(f"Optimal PCA components: {optimal_pca_components}")
    print(f"Optimal clusters: {optimal_k}")
    print(f"Silhouette Score: {silhouette_final:.4f}")
    print(f"Calinski-Harabasz Score: {ch_final:.4f}")
    print(f"Davies-Bouldin Score: {db_final:.4f}")
    print("Cluster distribution:")
    for cluster, count in cluster_distribution.items():
        print(f"  Cluster {cluster}: {count} samples")

    countries_by_cluster = {}
    for cluster in np.unique(final_labels):
        cluster_indices = np.where(final_labels == cluster)[0]
        countries = df_category.iloc[cluster_indices]['Country'].tolist()
        countries_by_cluster[cluster] = countries

    print("Countries by cluster:")
    for cluster, countries in countries_by_cluster.items():
        print(f"  Cluster {cluster}: {countries}")

metrics_df = pd.DataFrame(all_metrics)
metrics_df.to_csv('category_clustering_metrics.csv', index=False)

plt.figure(figsize=(15, 10))
```

```python
categories_list = [m['Category'] for m in all_metrics]
silhouette_vals = [m['Silhouette'] for m in all_metrics]
ch_vals = [m['Calinski_Harabasz'] for m in all_metrics]
db_vals = [m['Davies_Bouldin'] for m in all_metrics]

x = np.arange(len(categories_list))
width = 0.25

plt.bar(x - width, silhouette_vals, width, label='Silhouette', alpha=0.8)
plt.bar(x, ch_vals, width, label='Calinski-Harabasz', alpha=0.8)
plt.bar(x + width, db_vals, width, label='Davies-Bouldin', alpha=0.8)

plt.xlabel('Category')
plt.ylabel('Score')
plt.title('Clustering Metrics by Category')
plt.xticks(x, categories_list, rotation=45)
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig('category_metrics_summary.png', dpi=300, bbox_inches='tight')
plt.show()

print("\n=== Summary Table ===")
print(metrics_df[['Category', 'Samples', 'Optimal_Clusters', 'Silhouette',␣
↪'Calinski_Harabasz', 'Davies_Bouldin']].to_string(index=False))

best_category = metrics_df.loc[metrics_df['Silhouette'].idxmax(), 'Category']
print(f"\nBest clustering by Silhouette score: {best_category}")
print(f"Summary saved to 'category_clustering_metrics.csv'")
```



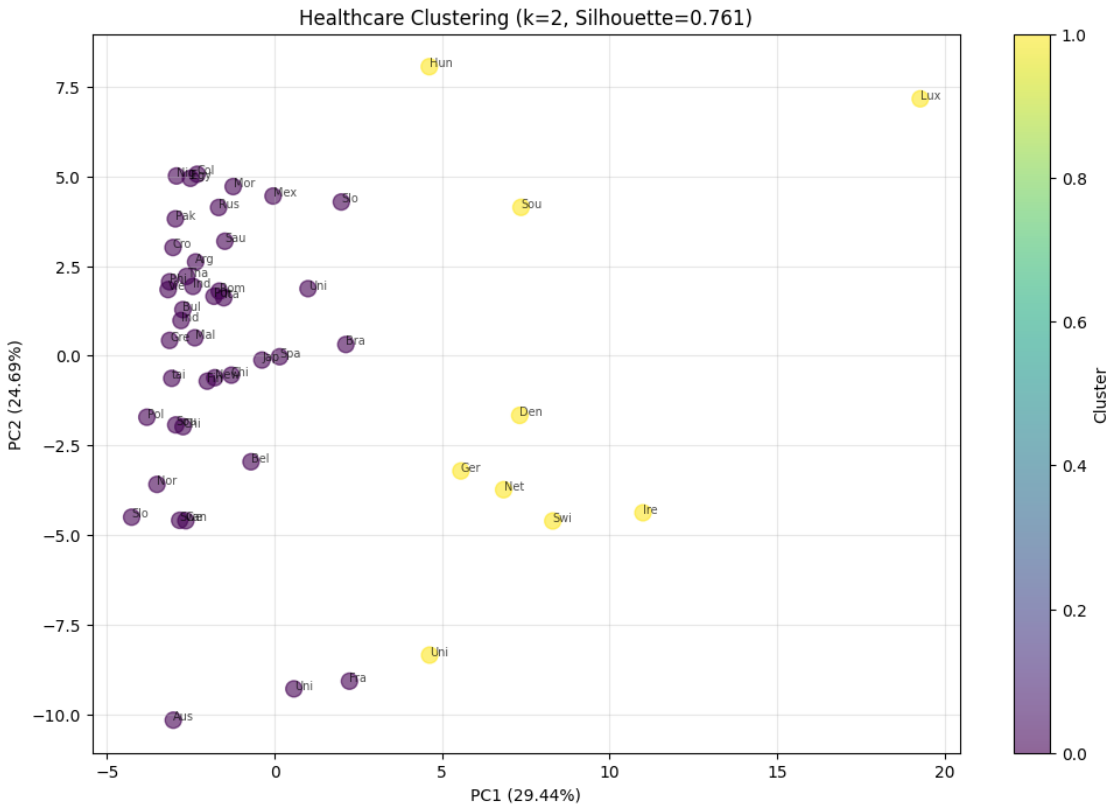

```
=== Consumer ===
Samples: 58
Optimal PCA components: 1
Optimal clusters: 2
Silhouette Score: 0.7258
Calinski-Harabasz Score: 120.1698
Davies-Bouldin Score: 0.4882
Cluster distribution:
  Cluster 0: 47 samples
  Cluster 1: 11 samples
Countries by cluster:
  Cluster 0: ['Argentina', 'Australia', 'Austria', 'Brazil', 'Bulgaria',
'Canada', 'Chile', 'China', 'Colombia', 'Croatia', 'Cyprus', 'Czechia',
'Denmark', 'Egypt', 'Estonia', 'Finland', 'Greece', 'Hungary', 'India',
'Indonesia', 'Italy', 'Japan', 'Kazakhstan', 'Latvia', 'Lithuania',
'Luxembourg', 'Malaysia', 'Malta', 'Morocco', 'New Zealand', 'Nigeria',
'Norway', 'Pakistan', 'Philippines', 'Poland', 'Romania', 'Russia', 'Saudi
Arabia', 'Slovakia', 'South Africa', 'South Korea', 'Sweden', 'taiwan',
'Thailand', 'United Arab Emirates', 'United Kingdom', 'Vietnam']
  Cluster 1: ['Belgium', 'France', 'Germany', 'Ireland', 'Mexico',
```

'Netherlands', 'Portugal', 'Slovenia', 'Spain', 'Switzerland', 'United States']

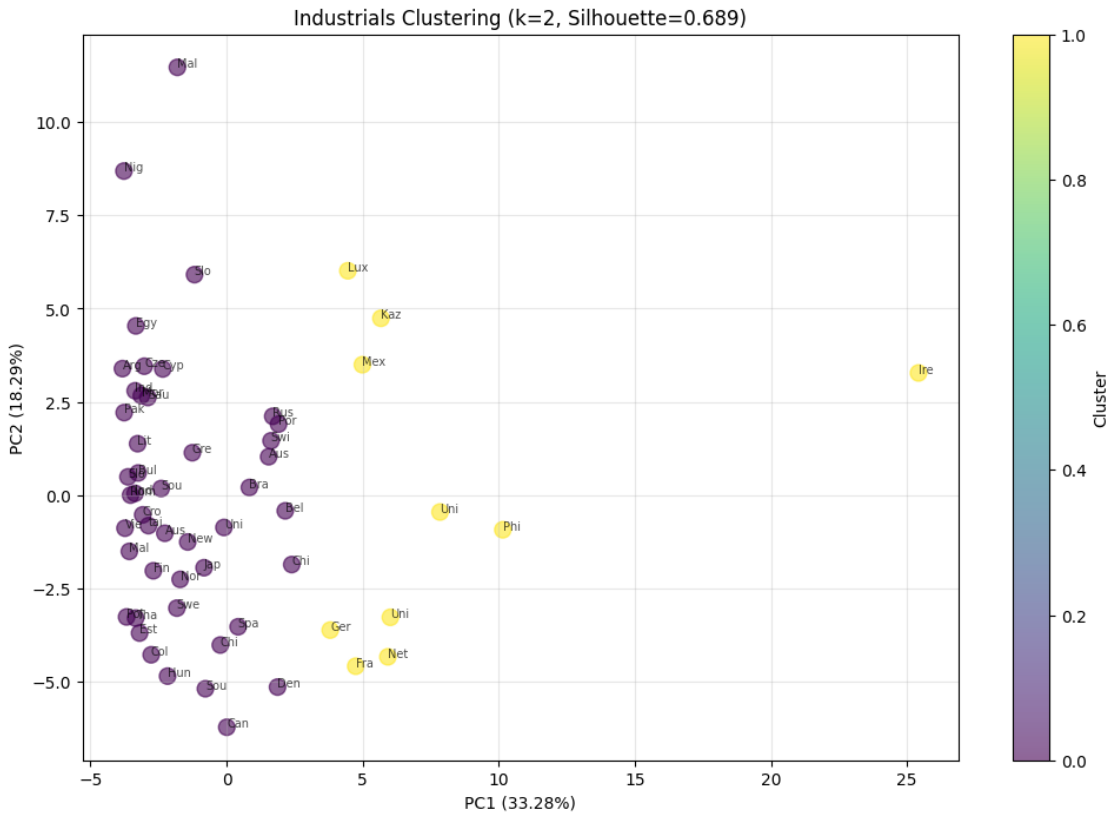



```
=== EnergyAndUtilities ===
Samples: 54
Optimal PCA components: 1
Optimal clusters: 2
Silhouette Score: 0.7817
Calinski-Harabasz Score: 124.9898
Davies-Bouldin Score: 0.5364
Cluster distribution:
  Cluster 0: 47 samples
  Cluster 1: 7 samples
Countries by cluster:
  Cluster 0: ['Argentina', 'Australia', 'Brazil', 'Bulgaria', 'Canada', 'Chile',
'China', 'Colombia', 'Croatia', 'Cyprus', 'Egypt', 'Estonia', 'France',
'Germany', 'Greece', 'Hungary', 'India', 'Indonesia', 'Italy', 'Japan',
'Kazakhstan', 'Lithuania', 'Luxembourg', 'Malaysia', 'Malta', 'Mexico',
'Morocco', 'Netherlands', 'New Zealand', 'Nigeria', 'Norway', 'Pakistan',
'Philippines', 'Poland', 'Romania', 'Russia', 'Slovenia', 'South Africa', 'South
Korea', 'Sweden', 'Switzerland', 'taiwan', 'Thailand', 'United Arab Emirates',
'United Kingdom', 'United States', 'Vietnam']
```

```
  Cluster 1: ['Austria', 'Czechia', 'Denmark', 'Finland', 'Portugal', 'Saudi
Arabia', 'Spain']
```

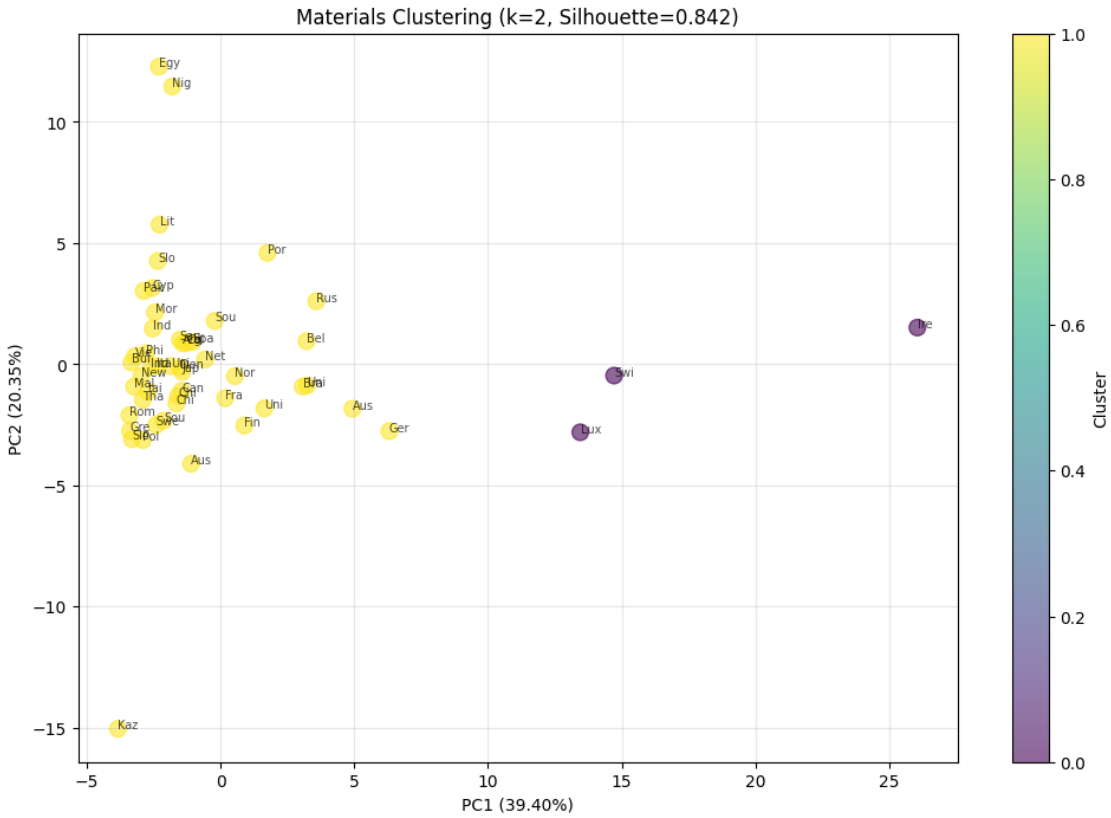

Healthcare Clustering (k=2, Silhouette=0.761)
PC2 (24.69%)
PC1 (29.44%)
Cluster
Hun
Lux
Sou
Den
Ger
Net
Swi
Ire
Bel
Bra
Uni
Fra
Aus
Mor
Mex
Rus
Sau
Arg
Mal
Nor
Pol
Bul
Tai
Pak
Gro
Slo
Spa
Can
Ind
Chi
Col
Ven
Rom
Ita
Jap
Vie
Ind
Ind
Ind

```
  Cluster 1: ['Denmark', 'Germany', 'Hungary', 'Ireland', 'Luxembourg',
'Netherlands', 'South Africa', 'Switzerland', 'United States']
```



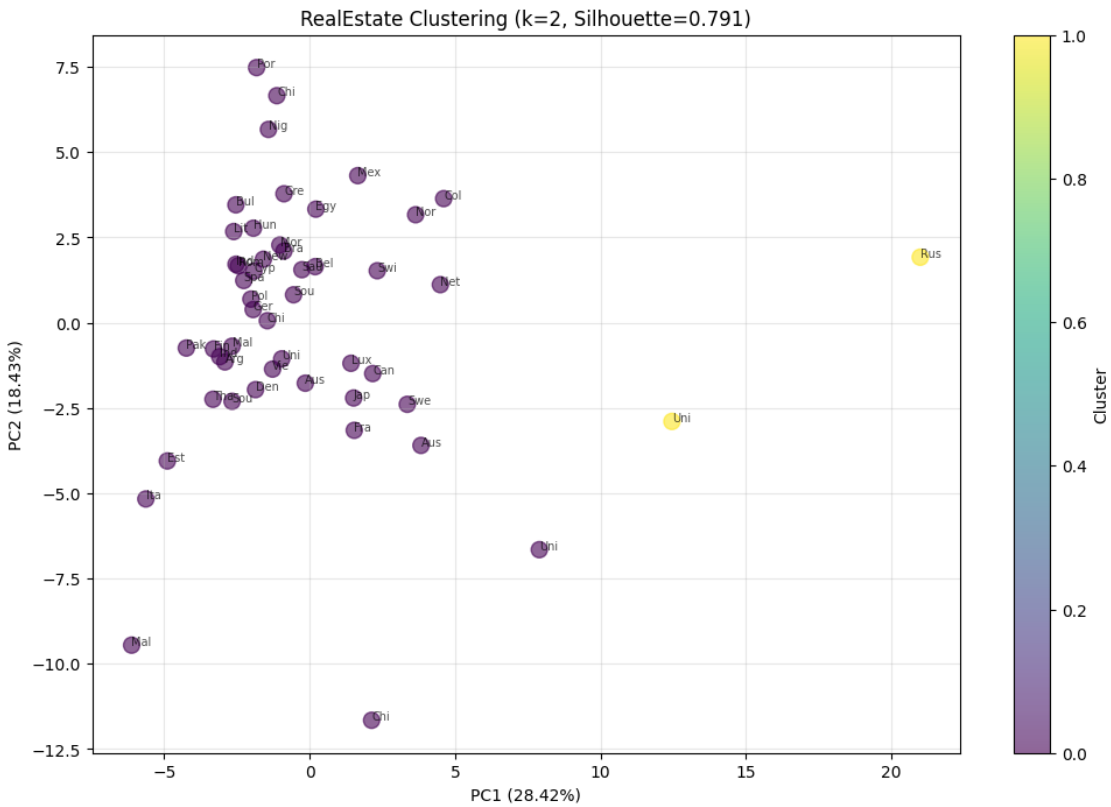

```
=== Industrials ===
Samples: 56
Optimal PCA components: 1
Optimal clusters: 2
Silhouette Score: 0.6885
Calinski-Harabasz Score: 75.3526
Davies-Bouldin Score: 0.5821
Cluster distribution:
  Cluster 0: 46 samples
  Cluster 1: 10 samples
Countries by cluster:
  Cluster 0: ['Argentina', 'Australia', 'Austria', 'Belgium', 'Brazil',
'Bulgaria', 'Canada', 'Chile', 'China', 'Colombia', 'Croatia', 'Cyprus',
'Czechia', 'Denmark', 'Egypt', 'Estonia', 'Finland', 'Greece', 'Hungary',
'India', 'Indonesia', 'Japan', 'Lithuania', 'Malaysia', 'Malta', 'Morocco', 'New
Zealand', 'Nigeria', 'Norway', 'Pakistan', 'Poland', 'Portugal', 'Romania',
'Russia', 'Saudi Arabia', 'Slovakia', 'Slovenia', 'South Africa', 'South Korea',
'Spain', 'Sweden', 'Switzerland', 'taiwan', 'Thailand', 'United Kingdom',
```

```
'Vietnam']
  Cluster 1: ['France', 'Germany', 'Ireland', 'Kazakhstan', 'Luxembourg',
'Mexico', 'Netherlands', 'Philippines', 'United Arab Emirates', 'United States']
```



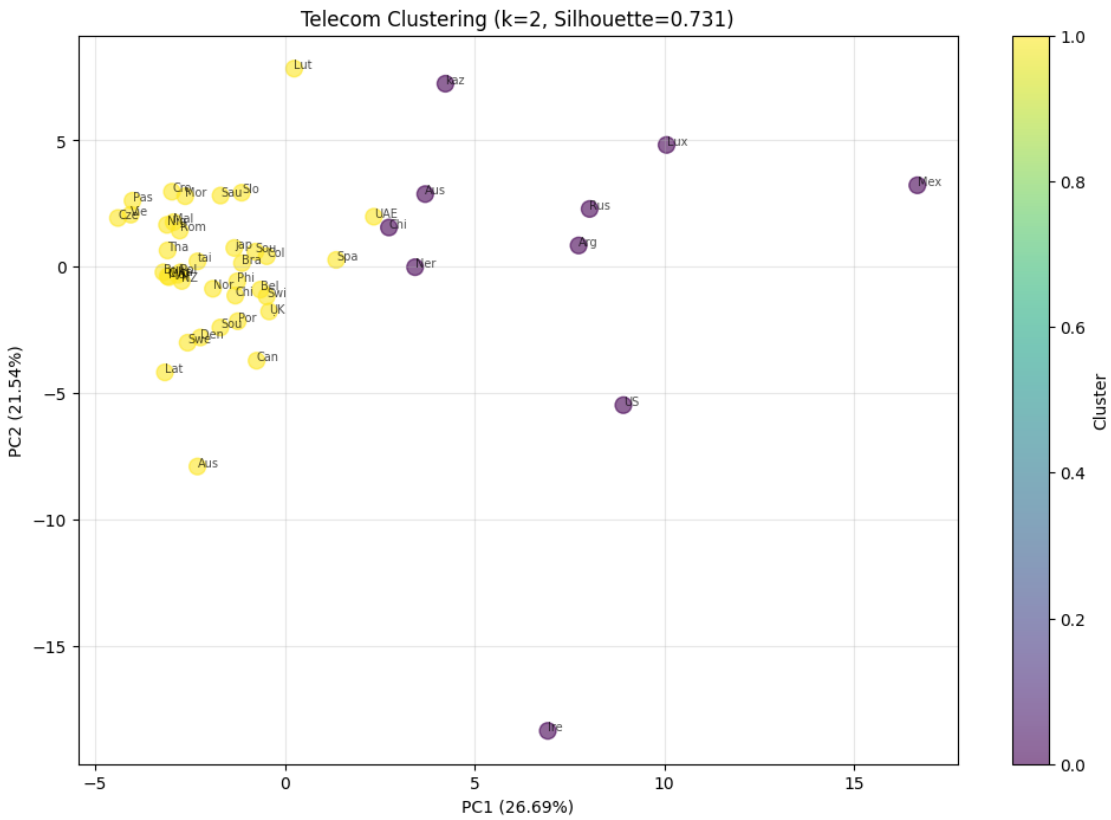

```
=== Materials ===
Samples: 51
Optimal PCA components: 1
Optimal clusters: 2
Silhouette Score: 0.8416
Calinski-Harabasz Score: 139.9898
Davies-Bouldin Score: 0.3710
Cluster distribution:
  Cluster 0: 3 samples
  Cluster 1: 48 samples
Countries by cluster:
  Cluster 0: ['Ireland', 'Luxembourg', 'Switzerland']
  Cluster 1: ['Argentina', 'Australia', 'Austria', 'Belgium', 'Brazil',
'Bulgaria', 'Canada', 'Chile', 'China', 'Colombia', 'Cyprus', 'Denmark',
'Egypt', 'Finland', 'France', 'Germany', 'Greece', 'India', 'Indonesia',
'Italy', 'Japan', 'Kazakhstan', 'Lithuania', 'Malaysia', 'Morocco',
'Netherlands', 'New Zealand', 'Nigeria', 'Norway', 'Pakistan', 'Philippines',
```

'Poland', 'Portugal', 'Romania', 'Russia', 'Saudi Arabia', 'Slovakia', 'Slovenia', 'South Africa', 'South Korea', 'Spain', 'Sweden', 'taiwan', 'Thailand', 'United Arab Emirates', 'United Kingdom', 'United States', 'Vietnam']



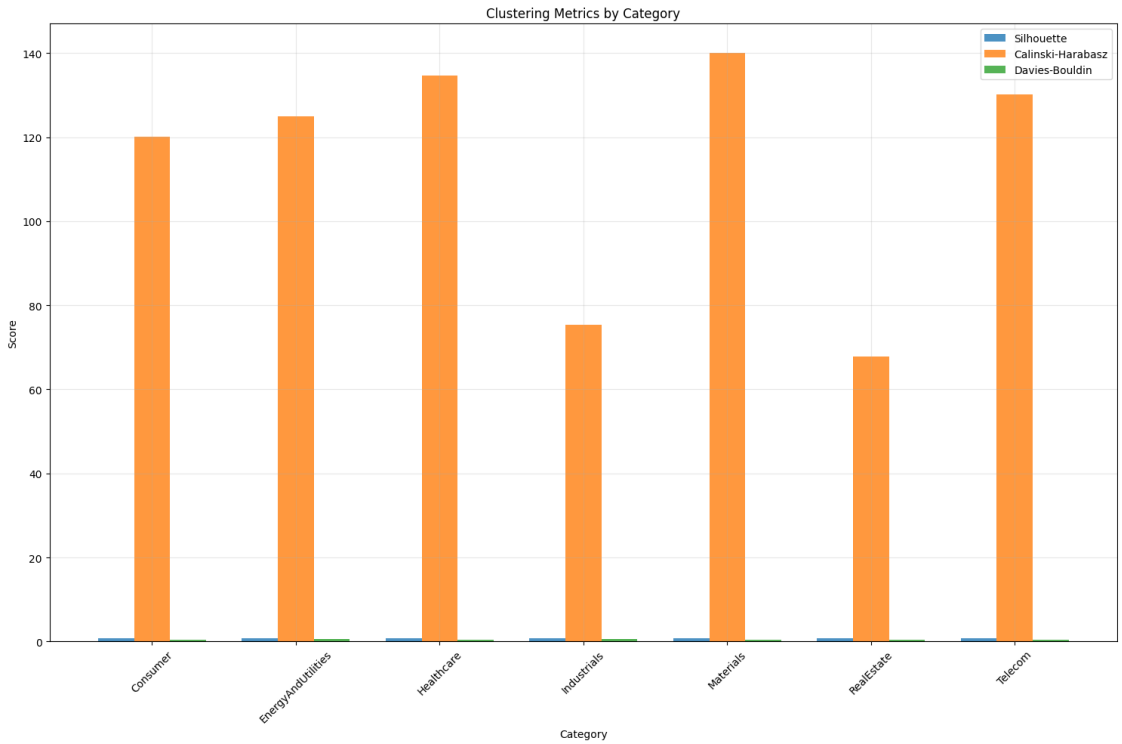

```
=== RealEstate ===
Samples: 50
Optimal PCA components: 1
Optimal clusters: 2
Silhouette Score: 0.7914
Calinski-Harabasz Score: 67.8685
Davies-Bouldin Score: 0.3707
Cluster distribution:
  Cluster 0: 48 samples
  Cluster 1: 2 samples
Countries by cluster:
  Cluster 0: ['Argentina', 'Australia', 'Austria', 'Belgium', 'Brazil',
'Bulgaria', 'Canada', 'Chile', 'China', 'Chinese Taipei', 'Colombia', 'Cyprus',
'Denmark', 'Egypt', 'Estonia', 'Finland', 'France', 'Germany', 'Greece',
'Hungary', 'India', 'Indonesia', 'Italy', 'Japan', 'Lithuania', 'Luxembourg',
'Malaysia', 'Malta', 'Mexico', 'Morocco', 'Netherlands', 'New Zealand',
```

```
'Nigeria', 'Norway', 'Pakistan', 'Poland', 'Portugal', 'Romania', 'Saudi
Arabia', 'South Africa', 'South Korea', 'Spain', 'Sweden', 'Switzerland',
'Thailand', 'United Kingdom', 'United States', 'Vietnam']
  Cluster 1: ['Russia', 'United Arab Emirates']
```




```
=== Telecom ===
Samples: 48
Optimal PCA components: 1
Optimal clusters: 2
Silhouette Score: 0.7308
Calinski-Harabasz Score: 130.1998
Davies-Bouldin Score: 0.4587
Cluster distribution:
  Cluster 0: 10 samples
  Cluster 1: 38 samples
Countries by cluster:
  Cluster 0: ['Argentina', 'Austria', 'Chile', 'Ireland', 'kazartan', 'Luxemb',
'Mexico', 'Nertheland', 'Russia', 'US']
  Cluster 1: ['Australia', 'Belgium', 'Brazil', 'Bulgatia', 'Canada', 'China',
'Colombia', 'Croatia', 'Cyprus', 'Czechia', 'Denmark', 'Italy', 'japan',
'Latvia', 'Luthunia', 'Malay', 'Malta', 'Moroco', 'Nigieria', 'Norway', 'NZ',
```

```
'Paskistan', 'Phil', 'Poland', 'Portuga', 'Romania', 'Saudi', 'Slovenia', 'South
Af', 'South Korea', 'Spain', 'Sweden', 'Swit', 'taiwan', 'Thailand', 'UAE',
'Vietnam', 'UK']
```




```
=== Summary Table ===
          Category  Samples  Optimal_Clusters  Silhouette  Calinski_Harabasz
Davies_Bouldin
          Consumer       58                 2    0.725804         120.169759
0.488163
EnergyAndUtilities       54                 2    0.781725         124.989805
0.536434
        Healthcare       50                 2    0.761441         134.697542
0.421255
       Industrials       56                 2    0.688504          75.352605
0.582113
         Materials       51                 2    0.841614         139.989769
0.371031
        RealEstate       50                 2    0.791395          67.868499
0.370661
           Telecom       48                 2    0.730773         130.199849
0.458735

Best clustering by Silhouette score: Materials
```

```
Summary saved to 'category_clustering_metrics.csv'
```